



APOLLO 13 Guide

PROJECT APOLLO - NASSP
ORBITER 2024

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TLI+90 PAD Calculation

Initialize MPT

RTCC Configuration page > Set CSM: Odyssey

RTCC Configuration page > Set CSM: Odyssey



BCK > MPT Initialization page > MED M55 > AUT > UPD

MED M49 > AUT > UPD

MED M50 > AUT > UPD

MED M51 > AUT > UPD

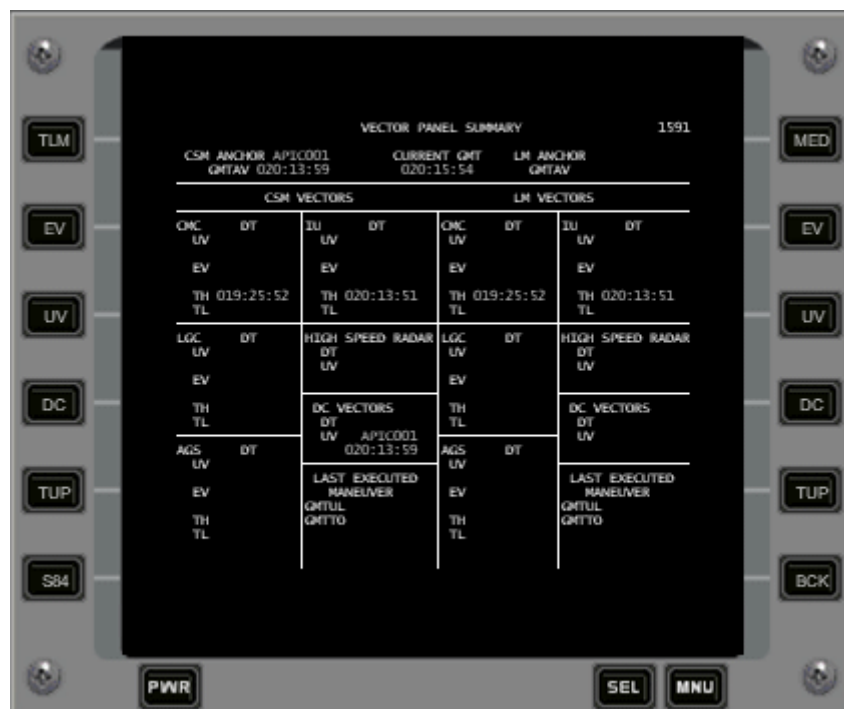


MPT Initialization page > VPS (Vector Panel Summary Page) > TLM

> DC > Odyssey

> DC > Odyssey

> TUP > DC



Add TLI Maneuver to MPT

Mission Plan Table Page > ACT (Enables MPT)

> TLI > M68,CSM,1;



Add Undocking maneuver to MPT

Mission Plan Table Page > INP > TIG > 3:06:0 (CSM/SIVB Sep Time)

> PAG > DOC (DOC: Undocking, CFG: C)

> PAG > CLC



The screenshot shows a terminal window titled "Active" with the "FDO MISSION PLAN TABLE" displayed. The table contains mission data for CSM STA ID APIC001 and LM STA ID GETAV. The table has columns for GETI, DT, DV, DVRM, HA, HP, and CODE. The data is as follows:

GETI	DT	DV	DVRM	HA	HP	CODE
002:35:40.8	000:30	10414.1	867	99999.9	121.8	CSJG01TL
003:06:00.0		0.0	0175	99999.9	122.7	CCSE02UD

The screen is surrounded by various buttons: ACT, INP, TU, POI, P13, P14, BCK, PWR, SEL, MNU, UDT, DEL, P16, P17.

Go back to MPT Page and check if Undocking maneuver was added



This screenshot is identical to the one above, showing the "FDO MISSION PLAN TABLE" with the same mission data and navigation buttons.

GETI	DT	DV	DVRM	HA	HP	CODE
002:35:40.8	000:30	10414.1	867	99999.9	121.8	CSJG01TL
003:06:00.0		0.0	0175	99999.9	122.7	CCSE02UD

Calculate TLI+90 Maneuver

BCK > TAR > ENT > AST (Abort Scan Table Inputs Page) > SIT > AOL (Splashdown Site)
VTI > 3:30:0 (CSM Vector Time)



AST (Abort Scan Table Inputs Page) > AST > CLC



Abort Scan Table > RTE (Return To Earth Digital Inputs page) > ULL > 4 0 (No Ullage)



RTE (Return To Earth Digital Inputs page) > DIS > CLC



DIS (Return To Earth Digital page) > TRA > M74,CSM,,RTEP; (Transfers TLI+90 Maneuver to MPT)

Go BCK to MPT page and check if TLI+90 maneuver is added

Active

FDO MISSION PLAN TABLE 0047

CSM STA ID APIC001 LM STA ID
GETAV 01:00:59 GETAV

GETI	DT	DV	DVRM	HA	HP	CODE
002:35:40.8	000:30	0414.1	867	99999.9	121.8	CSJG01TL
003:06:00.0	000:55	0.0	0175	99999.9	122.7	CCSE02UD
004:00:47.0		6371.3	3812	21078.3	18.8	CCSE03RE

Obtain TLI+90 Maneuver PAD

BCK > PAD > MAN (Maneuver PAD page) > CLC > CSM 3 (Maneuver n#3 in MPT)
> ULL > 4 0 (No Ullage)



P37 L/O+8 PAD Calculation

Delete TLI+90 Maneuver from MPT

MPT Page > DEL > M62,CSM,3,D;; (Deletes MPT maneuver n#3, which is TLI+90)

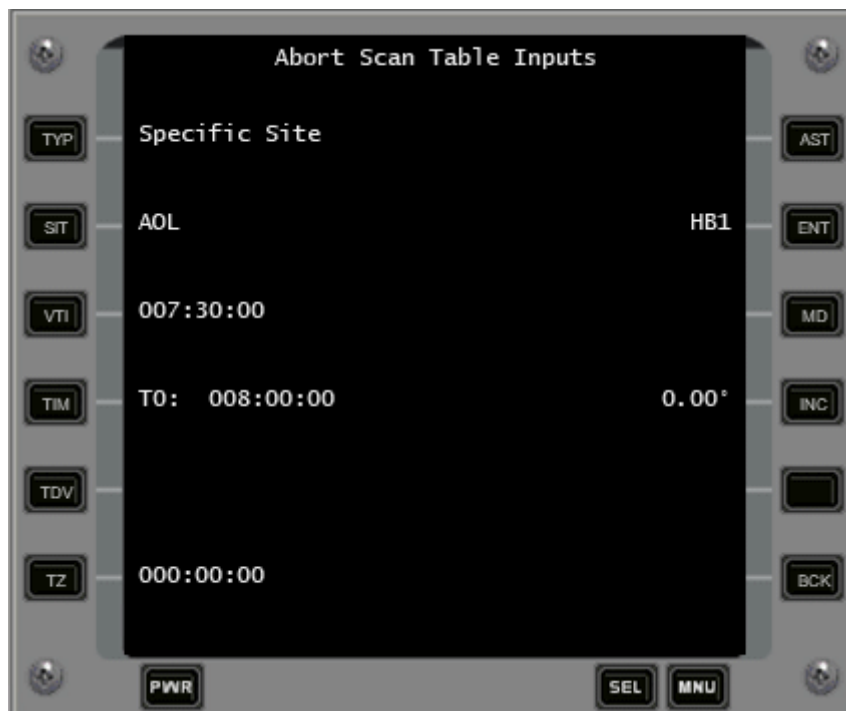


Calculate L/O+8 Maneuver

TAR > ENT > AST (Abort Scan Table Inputs Page) > SIT > MPL (Splashdown Site)

VTI > 7:30:0 (CSM Vector Time)

TIM > 8:00:0 (L/O+8 TIG)



AST (Abort Scan Table Inputs Page) > AST > CLC

(P37 Maneuvers doesn't require a P30 PAD. Take screenshot of this page)

ABORT SCAN TABLE 0362

CODE STA ID	SITE AM	GETI GETV	DV INCL HPC	VEI GEI	GETEI GETL	LAT IP LNG IP
103	ACL EDATP	004:05:00 003:26:05	6373 A31.80 0.0	33996 -6.21	013:35:34 013:43:48	22:20S 025:00W
105	MPL EDATP	008:00:00 007:19:59	7819 A31.79 0.0	34958 -6.35	022:35:36 022:43:51	24:05S 165:00W

TLI PAD Calculation

Disable MPT and delete all Maneuvers

MPT Page > ACT (Sets MPT to Inactive)

> DEL > M62,CSM,1,D,; (Deletes all Maneuvers)

Inactive

FDO MISSION PLAN TABLE 0047

CSM STA ID APIC001
GETAV 01:00:59

LM STA ID
GETAV

GETI	DT	DV	DVRM	HA	HP	CODE
------	----	----	------	----	----	------

Obtain TLI PAD

PAD > MAN > OPT > TLI PAD > CLC

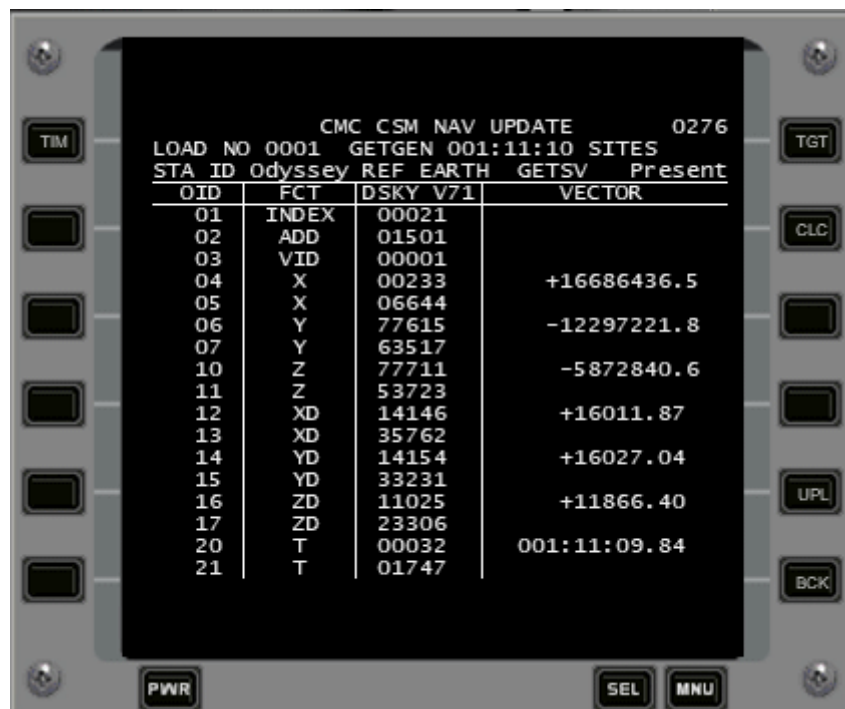


Uplink CSM State Vector & V66

Check if **MPT is Inactive** otherwise disable it

UPL (Uplinks page) > DIS > 0 (CSM Nav Update) > TGT > Odyssey > CLC > UPL

Before Uplink check P00 and UPTLM CM Switch is in ACCEPT position



After the Uplink do a V66 to transfer the CSM vector to LM Slot

Uplink SIVB Yaw Maneuver

Project Apollo MFD > IU > SRC > AS-508-S4BSTG (Define SIVB Vehicle)

> TYP > Evasive Yaw Mnvr Enable > UPL

Verify Status: Uplink Accepted



Uplink SIVB Evasive Maneuver

Project Apollo MFD > IU > TYP > Timebase 8 Enable > UPL

Verify Status: Uplink Accepted



Uplink PTC REFSMMAT

Calculate PTC REFSMMAT

RTCC MFD > UTI (Utility Page) > REFSMMAT > CSM: Odyssey
> OPT > PTC > CLC



Uplink REFSMMAT

BCK > UPL (Uplinks page) > DIS > 12 (REFSMMAT Update) > CLC > CUR 2 (Desired REFSMMAT)



P37 L/O+15 PAD Calculation

Calculate L/O+15 Maneuver

TAR > ENT > AST (Abort Scan Table Inputs Page) > SIT > MPL (Splashdown Site)

TIM > 15:00:0 (L/O+15 TIG)



AST (Abort Scan Table Inputs Page) > AST > CLC

(P37 Maneuvers doesn't require a P30 PAD. Take screenshot of this page)

ABORT SCAN TABLE 0362

CODE STA ID	SITE AM	GETI GETV	DV INCL HPC	VEI GEI	GETEI GETL	LAT IP LNG IP	
101	MPL EDATP	015:00:00 005:34:35	5607 A31.81 0.0	35536 -6.43	047:04:17 047:12:34	20:53S 165:00W	

Uplink SIVB APS MCC-1 & MCC-2

Calculate SIVB APS MCC-1

RTCC MFD > UTI > LUN (Lunar Impact Page) > S4B > IU: AS-508-S4BSTG

> SET > MOD: Calculate Burn

> TIG: 6:0:0 (APS MCC-1 TIG)

> LAT: -2.60 (Latitude of S4B Impact)

> LON: -27.70 (Longitude of S4B Impact)

> OPT: Optimize dV

RTCC MFD > UTI > LUN (Lunar Impact Page) > CLC



Copy Burn Data TIG, BT, Pitch and Yaw

Uplink SIVB APS MCC-1 with PAMFD

Project Apollo MFD > IU > TYP > S-IVB/IU Lunar Impact

> IN1 > 6110 (RTCC TIG Data)

> IN2 > Load RTCC Burn Time

> IN3 > Load RTCC Pitch

> IN4 > Load RTCC Yaw

IU > TYP > UPL

Verify Status: Uplink Accepted



For APS MCC-2 Calculation change TIG to 9:0:0 (9h)

MCC-1 Calculation

Define LM

Go to RTCC MFD Configuration page and set Aquarius as LM

Load Preflight Skeleton Flight Plan Table

RTCC MFD > TAR > TL > SFP > F62 > F62,,1,72.0;

SKELETON FLIGHT PLAN TABLE		1597
BLOCK NUMBER	1 (PREFLIGHT)	
01-GMT TIME FLAG	028:39:55.82	
02-MODE	0	
03-GMT OF TLI PERICYN.	096:53:22.00	
04-LAT. OF TLI PERICYN.	-0.058	
05-LONG. OF TLI PERICYN.	180.155	
06-HEIGHT OF TLI PERICYN.	211.700	
07-GMT OF LOI PERICYN.	096:41:39.00	
08-LAT. OF LOI PERICYN.	-0.343	
09-LONG. OF LOI PERICYN.	177.996	
10-HEIGHT OF LOI PERICYN.	57.000	
11-GMT OF TLI IGNITION	002:35:24.00	
12-GMT OF NODE	096:41:39.00	
13-LAT. OF NODE	-0.343	
14-LONG. OF NODE	177.996	
15-HEIGHT OF NODE	57.000	
16-DELTA AZIMUTH OF LOI	-0.700	
17-FLIGHT PATH ANGLE AT LOI	2.000	
18-DELTA TIME OF LPO	090:00:00.00	
19-DELTA TIME OF LLS	026:13:35.00	
20-AZIMUTH OF LLS	-93.880	
21-LAT. OF LLS	-3.669	
22-LONG. OF LLS	-17.484	
23-RADIUS OF LLS	937.734	
24-DELTA AZIMUTH OF TEI	2.000	
25-DELTA V OF TEI	3449.500	
26-DELTA TIME OF TEI	072:00:00.00	

Constraint LOI

Go BCK to Translunar Options page > CON > F23 > 77:24:53,77:26:53 (FP LOI Time is 77:24:53)



MCC-1 Calculation

BCK to Translunar Options page > MCC (MCC Planning page) > MAN > Option 4

> TIG > 11:41:25 (MCC1 TIG)



Midcourse Correction Planning page > MID > CLC

Tweak F23 (TLMIN/MAX) on MCC Inputs and Constants page until GET LOI on Midcourse Tradeoff corresponds to Flight Plan LOI TIG (77:24:53)

COLUMN	1	2	3	4	5	6
MODE	4					
RETURN	NONFREE					
AZ MIN	-93.880					
AZ MAX	-93.880					
WEIGHT	97129					
GETMCC	011:41					
DV MCC	20.7					
YAW MCC	-8.195					
H PYCN	60.859					
GET LOI	077:24					
DV LOI	2801.4					
AZ ACT	-93.880					
I FR	0.000					
I PR	40.593					
V EI	36135.9					
G EI	-6.520					
GETTEI	169:14					
DV TEI	2943.0					
DV REM	0.0					
GET IC	241:45					
LAT IP	-1.394					
LNG IP	189.969					
DV PC	94.6					

MCC-2 Calculation

Go BCK to MCC Planning page, change TIG to 30:40:49 and COL to 2

Midcourse Correction Planning	
MAN	Option 4: Non-FR BAP, Fixed LPO, LS
VTI	CSM: Odyssey
TIG	030:40:49 GET
COL	2
CFG	Docked
SFP	1 (Preflight)

Midcourse Correction Planning page > MID > CLC

MODE	COLUMN 1	MIDCOURSE 2	TRADEOFF 3	4	5	0079 6
RETURN	NONFREE	NONFREE				
AZ MIN	-93.880	-93.880				
AZ MAX	-93.880	-93.880				
WEIGHT	97129	97129				
GETMCC	011:41	030:41				
DV MCC	20.7	32.5				
YAW MCC	-8.195	-7.056				
H PYCN	60.859	60.607				
GET LOI	077:24	077:24				
DV LOI	2801.4	2808.2				
AZ ACT	-93.880	-93.880				
I FR	0.000	0.000				
I PR	40.593	40.483				
V EI	36135.9	36133.8				
G EI	-6.520	-6.520				
GETTEI	168:14	168:49				
DV TEI	2943.0	2933.4				
DV REM	0.0	0.0				
GET LC	241:45	241:45				
LAT IP	-1.394	-1.436				
LNG IP	189.969	189.970				
DV PC	94.6	94.6				

MCC-1 Decision

Column #1 is MCC1, Column #2 is MCC2, compare both solutions

Scrub MCC-1 if the MCC-2 Propellant Cost is not prohibitive

Nominally MCC-1 should be scrubbed then burn MCC-2

P37 L/O+25, 35, 45 & 60 PAD Calculation

Calculate the same as L/O+15. TIM's are 25:00:00, 35:00:00, 45:00:00 & 60:00:00

CODE STA ID	SITE AM	GETI GETV	DV INCL HPC	VEI GEI	GETEI GETL	LAT IP	LNG IP
101	MPL	025:00:00	9940	36101	047:54:32	25:56S	
	EDATP	012:06:52	A31.83	-6.49	048:02:50	168:41E	
102	MPL	035:00:00	7704	36055	070:36:02	23:20S	
	EDATP	012:07:05	A31.89	-6.49	070:44:19	165:00W	
103	MPL	045:00:00	6176	36040	094:52:24	21:18S	
	EDATP	012:07:13	A32.03	-6.49	095:00:42	165:00W	
104	MPL	060:00:00	10100	36689	094:52:35	25:48S	
	EDATP	012:07:19	A32.52	-6.57	095:00:54	176:42W	

MCC-2 Calculation

Calculate the same as in MCC-1 Step

COLUMN	1	2	3	4	5	0079 6
MODE	4					
RETURN	NONFREE					
AZ MIN	-93.880					
AZ MAX	-93.880					
WEIGHT	97129					
GETMCC	030:41					
DV MCC	32.4					
YAW MCC	-7.025					
H PYCN	60.608					
GET LOI	077:24					
DV LOI	2808.1					
AZ ACT	-93.880					
I FR	0.000					
I PR	40.483					
V EI	36133.8					
G EI	-6.520					
GETTEI	168:49					
DV TEI	2933.4					
DV REM	0.0					
GET LC	241:45					
LAT IP	-1.436					
LNG IP	189.970					
DV PC	94.6					

Midcourse Tradeoff page > F30 > F30,1(MCC-2 Column number);

Calculate MCC-2 PAD

Midcourse Tradeoff page > TRA > MCC Transfer page > NUM: MCC2 Column number

- > THR: SPS
- > ITE: Iterate
- > CLC



Go to Maneuver PAD page and CLC



Uplink Target Load

RTCC MFD > Uplinks page > DIS > 10 (CMC External DV) > CLC > UPL



The image shows a screen from the RTCC MFD Uplink Target Load menu. The screen displays a table of data for various sites. The table has four columns: OI, FCT, DSKY V71, and DECIMAL. The data is as follows:

OI	FCT	DSKY V71	DECIMAL
01	INDEX	00012	
02	ADD	03404	
03	VGX	77767	-20.7
04	VGX	75567	
05	VGX	00000	+2.5
06	VGX	37646	
07	VGZ	77766	-24.8
10	VGZ	52522	
11	TIGN	01242	030:40:46.11
12	TIGN	03403	

The screen also shows the title "CMC EXTERNAL DV UPDATE" and the date "0264". The load number is "LOAD NO 1001" and the time is "GETGEN 028:41:49". The screen is surrounded by buttons: TIG, DV, CLC, UPL, BCK, SEL, MNU, and PWR.

LOI-5hr Flyby Abort PAD

Calculate LOI-5 Maneuver

RTCC MFD > TAR > ENT > AST > TIM: 72:25:0 (GET LOI -5h)

> TZ: 167:0:0 (Splashdown GET)

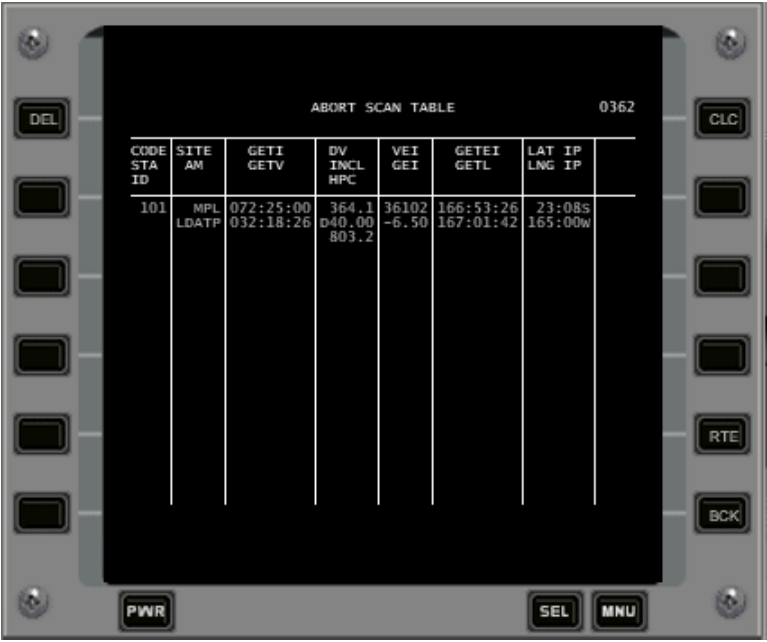


The image shows a screen from the RTCC MFD Abort Scan Table Inputs menu. The screen displays a form with the following fields:

- Specific Site
- MPL
- CSM: Odyssey
- T0: 072:25:00
- 167:00:00

The screen is surrounded by buttons: TYP, SIT, VTI, TIM, TDV, TZ, AST, ENT, MD, INC, BCK, SEL, MNU, and PWR.

Abort Scan Table Inputs page > AST > CLC



Abort Scan Table > RTE > AST: (Maneuver Code STA ID)

> ULL: 4 0 (No Ullage)

> DIS > CLC > TRA > RTEP



[Calculate LOI-5 Flyby PAD](#)

Go BCK to RTCC Main page > PAD > MAN > CLC

MCC-3 Calculation

RTCC MFD > TAR > TL > MCC > MAN > Option 1

> TIG > 55:24:53

> SFP 2 (Nominal Targets)



MCC Planning page > MID > CLC

MODE	COLUMN	1	2	3	4	5	6
RETURN		N/A					
AZ MIN		0.000					
AZ MAX		0.000					
WEIGHT		95779					
GETMCC		055:25					
DV MCC		0.5					
YAW MCC		-86.483					
H PYCN		60.942					
GET LOI		077:24					
DV LOI		2804.9					
AZ ACT		-93.880					
I FR		0.000					
I PR		0.000					
V EI		0.0					
G EI		0.000					
GETTEI		000:00					
DV TEI		0.0					
DV REM		0.0					
GET LC		000:00					
LAT IP		0.000					
LNG IP		0.000					
DV PC		0.0					

Nominally MCC-3 should be scrubbed, burn it if dV is more than 5ft/s

LM DAP PAD

RTCC MFD > PAD > DAP > VES: LM

> LM: Aquarius

> CLC



PC+2 (AOL) Abort PAD

Contraint RTE

RTCC MFD > TAR > ENT > CONT > Set VRMAX: 37500



Calculate PC+2 to AOL

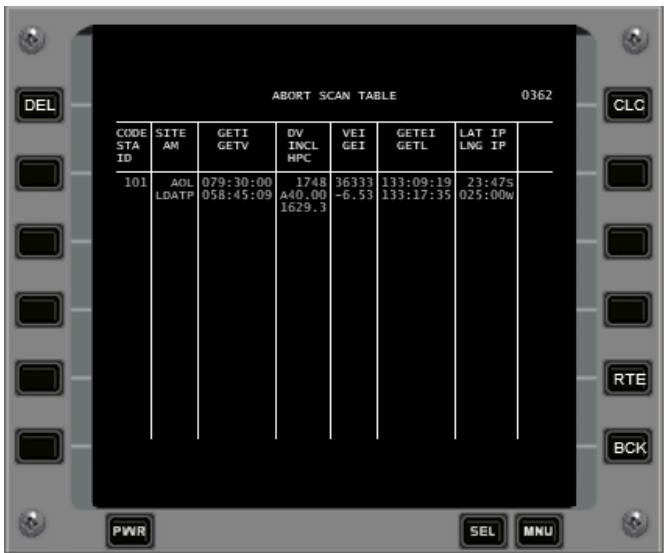
Assumes NO Free Return

BCK > AST > SIT: AOL

- > CSM: Aquarius
- > TIM: 79:30:0 (PC+2 TIG)
- > TZ: 133:0:0 (Splashdown GET)



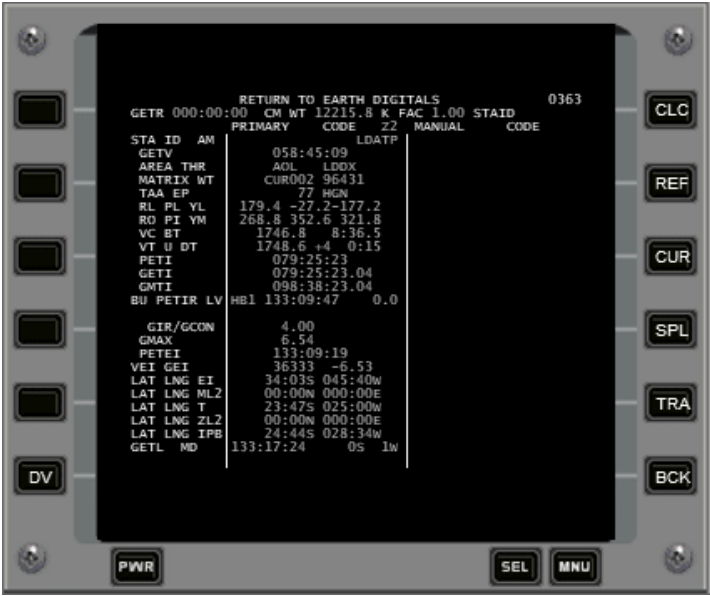
Abort Scan Table > CLC



Abort Scan Table > RTE > AST: PC+2 (AOL) Maneuver Code
> MAN: LDDX



RTE Digital Inputs > DIS > CLC



TRA > RTEP

Calculate P30 LM Maneuver PAD

LGC Clock Delta T

RCC MFD > UPL > DIS > 24 (LGC Time Increment) > CLC > INP (Enter Delta T)

Load **Increment** with **V55**



Uplink LM PTC REFSMMAT

RCC MFD > UTI > REF > VES: LM, LM: Aquarius

> OPT > PTC > CLC

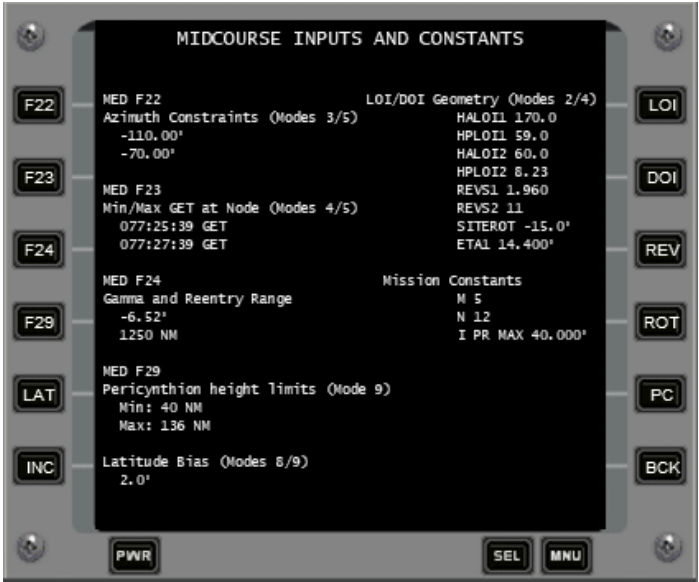
Go BCK to RTCC MFD Main page > UPL > DIS > 23 (LGC REFSMMAT Update) > CLC > CUR 1 (Actual)



MCC-Free Return PAD Calculation

Constraint Max HPC

RTCC MFD > TAR > TL > CON (MCC Constraints) > F29 > 40,136 (Set max HPC to 136 NM)



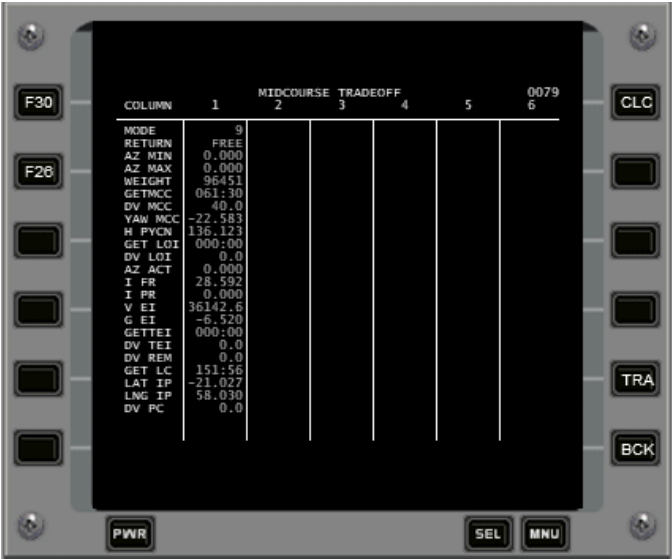
Calculate MCC-Free Return

BCK > MCC Processor > MAN: Option 9 (Fuel Critical Lunar Flyby)

- > CSM: Aquarius
- > TIG: 61:30:09
- > SFP: 2 (Nominal Targets)



MCC Planning page > MID > CLC



MCC Tradeoff page > TRA > TAB: LEM

- > NUM: (MCC Tradeoff Column Number)
- > THR: DPS
- > CLC



Obtain MCC-Free Return PAD

RTCC MFD > PAD > MAN > CLC P30 LM Maneuver PAD (Heads Down)



Calculate a LM DAP PAD to obtain GDA Trim Angles

PC+2 (MPL) Abort PAD

Contraint RTE

RTCC MFD > TAR > ENT > CONT > Set VRMAX: 37500

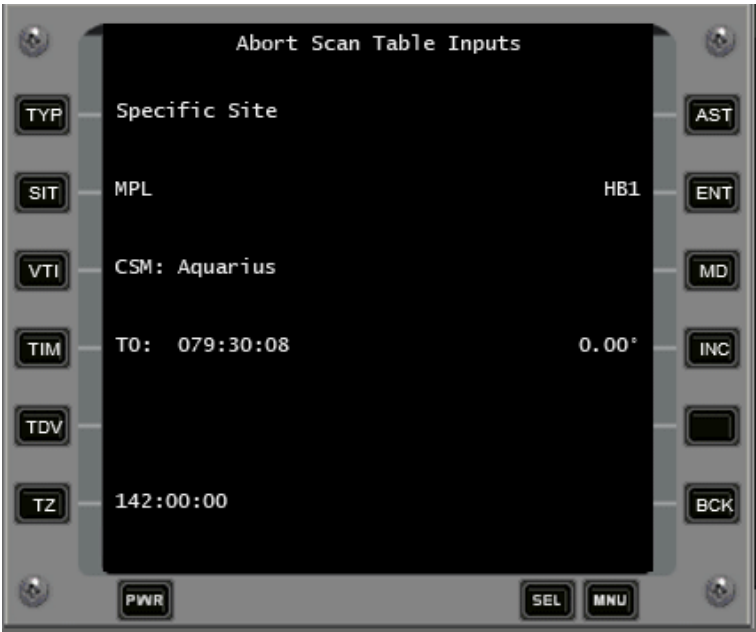


Calculate PC+2 to MPL

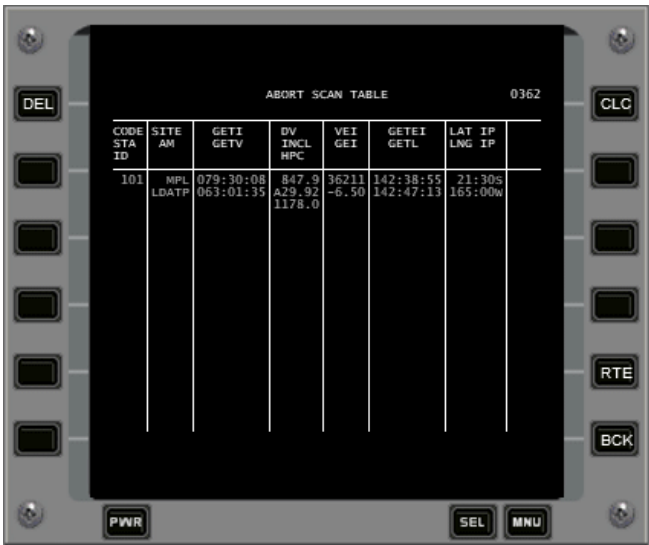
Assumes NO Free Return

BCK > AST > SIT: MPL

- > CSM: Aquarius
- > TIM: 79:30:8 (PC+2 TIG)
- > TZ: 142:0:0 (Splashdown GET)



Abort Scan Table > CLC



Abort Scan Table > RTE > AST: PC+2 (MPL) Maneuver Code

> MAN: LDDX

> ULL: 2 10 (2 Jets 10 sec)



RTE Digital Inputs > DIS > CLC

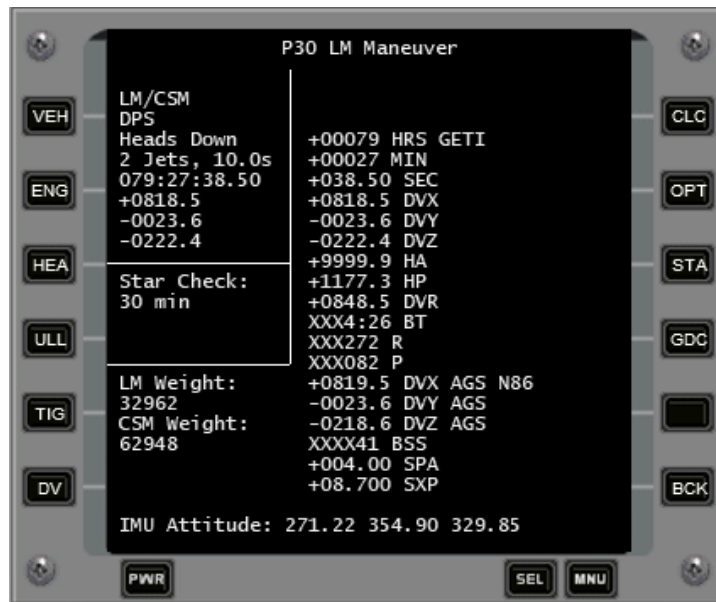


TRA > RTEP (Transfer Solution to PAD)

TRA > SPL > P (Save Splashdown site)

Obtain PC+2 (MPL) PAD

RTCC MFD > PAD > MAN > CLC P30 LM Maneuver PAD (Heads Down)



PC+2 (MPL) Target Load

RTCC MFD > UPL > DIS > 22 (External dV) > CLC > UPL

Map Update

RTCC MFD > PAD > MAP > GET > 76:0:0 > CLC



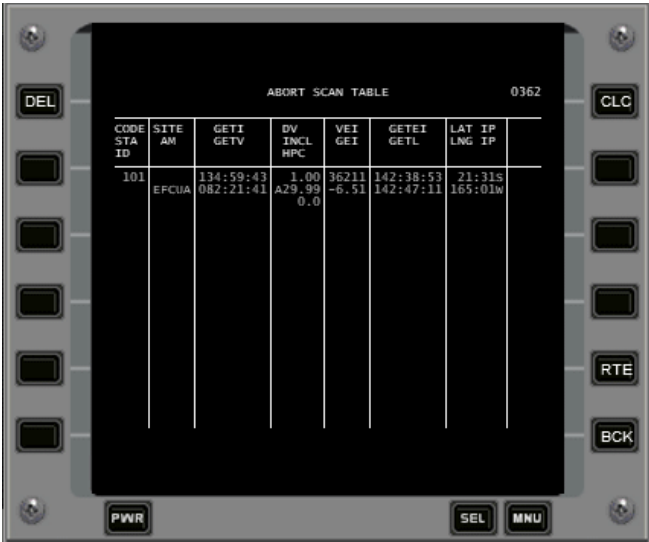
MCC-7 PAD (assumes NO MCC-5)

Calculate MCC-7 (no MCC-5 assumed)

RTCC MFD > TAR > ENT > AST > TYP: Unspecified Area, SIT: FCUA, CSM: Aquarius
> TIM: 134:59:43



Abort Scan Table Inputs > AST > CLC



RTE > AST: Maneuver Code STA ID, MAN: LDDX or LRDX (DPS or RCS Burn as required)

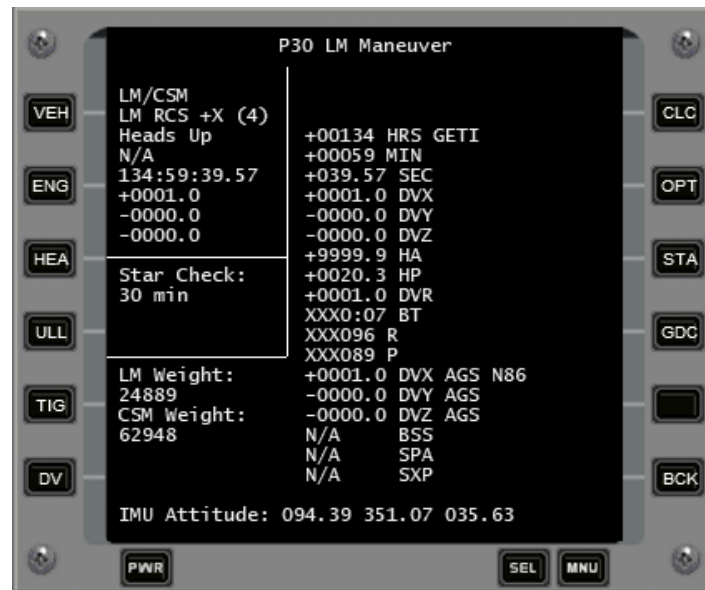


RTE Inputs page > DIS > CLC > TRA > RTEP (Transfer mnvr for PAD Calc)



Obtain MCC-7 (no MCC-5) PAD

RTCC MFD > PAD > MAN > Calculate P30 LM PAD



This is an **AGS Burn**, so just use dV values and Burn Time

Entry PAD (assumes NO MCC-5)

RTCC MFD > PAD > ENP > OPT > Lunar Entry PAD > CLC



MCC-5 PAD

In the Actual mission this maneuver was executed due a **venting problem** in the LM that messed up the trajectory (**which is not simulated in NASSP**). You should **mess up the trajectory** (overburning PC+2 or using TTCA) until GEI $\approx$ -8.8 degrees to get a MCC-5 solution close to the IRL

To **see GEI** go to RTCC MFD > MCC > MSK > 82 > GET > 3 105:0:0 and look at GEI value in column 3

Calculate MCC-5

RTCC MFD > TAR > ENT > AST > TYP: Unspecified Area, SIT: FCUA, CSM: Aquarius
> TIM: 105:18:28



Abort Scan Table Inputs page> AST > CLC

ABORT SCAN TABLE 0362

CODE STA ID	SITE AM	GETI GETV	DV INCL MPC	VEI GEI	GETEI GETL	LAT IP	LNG IP
101	EFCUA	105:18:28 082:22:27	3.88 A30.00 0.0	36211 -6.51	142:37:46 142:46:05	21:31S	164:45W

DEL CLC

PWR SEL MNU

If you messed up correctly your GEI, will obtain a dV between 4/8 fps

RTE > AST: Maneuver Code STA ID, MAN: LDDX, ULL: 4 10



RTE Inputs page > DIS > CLC > TRA > RTEP (Transfer mnvr for PAD Calc)



Obtain MCC-5 PAD

RTCC MFD > PAD > MAN > Calculate P30 LM PAD



This is an **AGS Burn**, so just use dV values and Burn Time

MCC-7 PAD Calculation

Calculate MCC-7

RTCC MFD > TAR > ENT > AST > SIT > Unspecified Area, FCUA

> CSM: Aquarius

> TIM: EI-5 hours



AST > CLC

CODE STA ID	SITE AM	GETI GETV	DV INCL HPC	VEI GETI	GETEI GETL	LAT IP LNG IP
105	EFCUA	137:39:00 110:56:47	5.17 A30.00 0.0	36211 -6.29	142:38:02 142:46:20	21:21S 164:20W

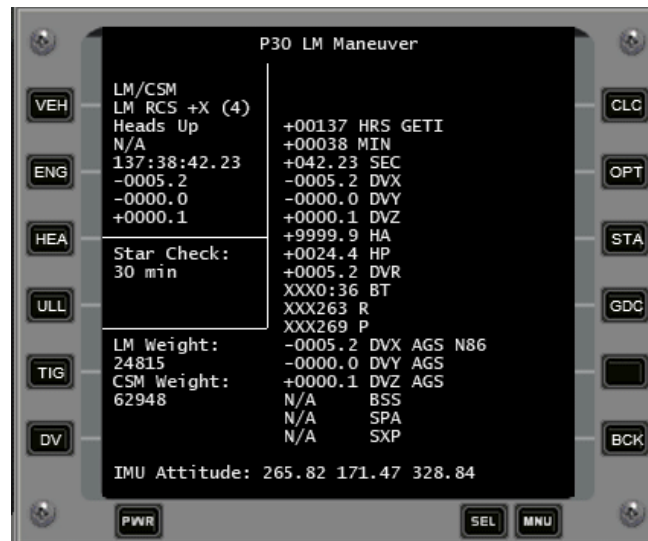
RTE > AST: Maneuver Code STA ID, MAN: LRDX, ULL: 4 Jets +X

Return to Earth Digital Inputs
Primary
105
60.00°
CUR
Heads Up
LRDX
Iterate
RCS Burn: +4 quads
Compute trim gimbals

DIS > CLC > TRA > RTEP (Transfer maneuver for PAD Calculation

Obtain MCC-7 PAD

RTCC MFD > PAD > MAN > Calculate P30 LM PAD



This is an **AGS Burn**, so just use dV values and Burn Time

Charge CM Batteries editing .scn

Save scenario and open it with any Text Editor. Search <ELECTRIC> section

Charge Battery A

Edit Battery A line like this:

<BATTERY> BATTERY_A 5508000.0000 294

Charge Battery B

Edit Battery A line like this:

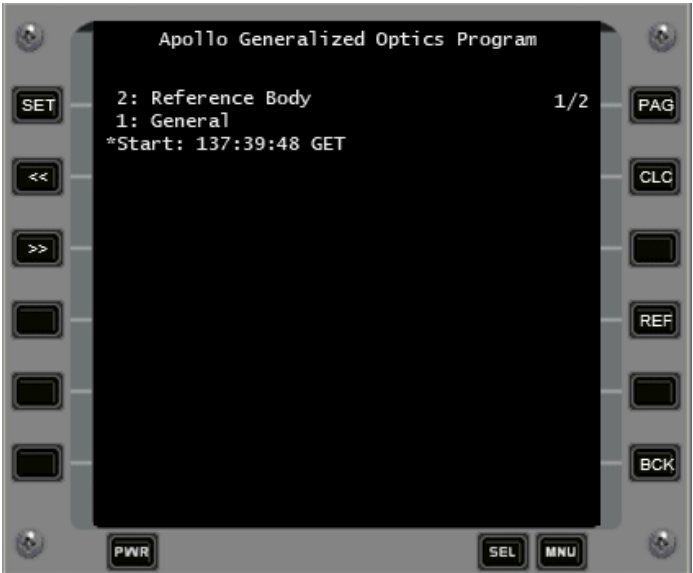
<BATTERY> BATTERY_B 5508000.0000 294

Crescent REFSMMAT (actual)

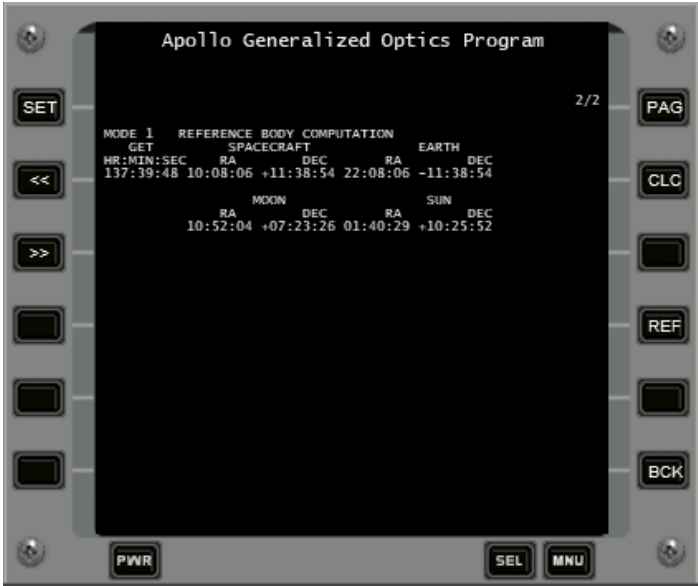
Calculate Earth and Sun Vectors

RTCC MFD > UTI > OPT > AGOP > SET > 2: Reference Body

> SET > Start: MCC-7 TIG



CLC > Go to Page 2



Copy Earth and Sun vectors (RA DEC)

Load Earth and Sun in Stars Slot

RTCC MFD > MCC > MSK > 1629 > MED: G13,400,(Earth RA),(Earth DEC);

> MED: G13,398,(Sun RA),(Sun DEC);



Calculate Crescent REFSMMAT

RTCC MFD > UTI > OPT > AGOP > SET > 2: Optical Support Table

> SET > 3: Compute REFSMMAT

> SET > Star 1: 400

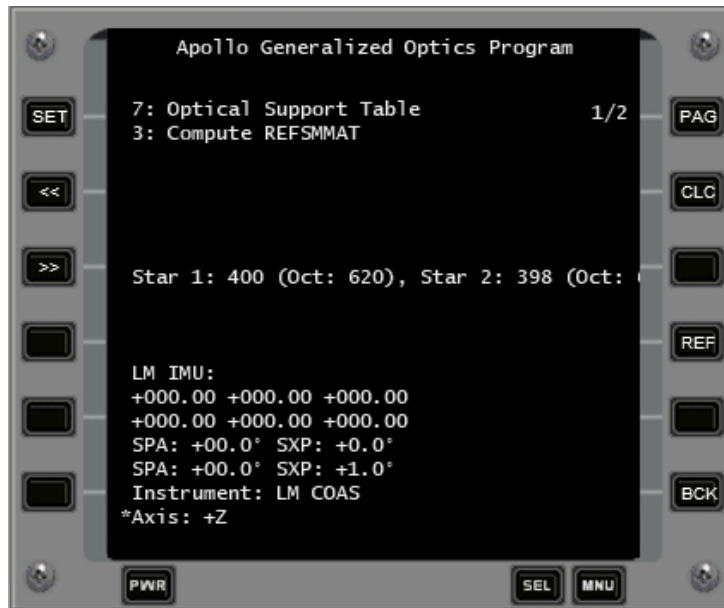
> SET > Star 2: 398

> SET > LM IMU

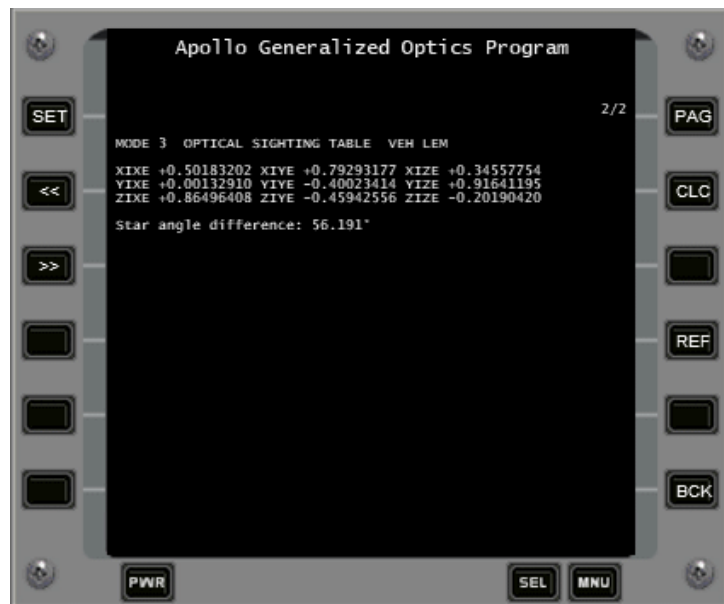
> SET > SFT (2nd): 0 +1

> SET > Instrument: LM COAS

> SET > Axis: +Z



CLC > Go to page 2 > REF (Computes new REFSMMAT)



RTCC MFD > UTI > REF > VES: LM, Aquarius > G00 > G00,LEM,OST,LEM,CUR;

Uplink Crescent REFSMMAT

RTCC MFD > UPL > DIS > 23 (12 For CSM) > CUR 1

MCC-7 REFSMMAT (desired)

Calculate MCC-7 REFSMMAT

First, calculate MCC-7 and transfer (RTEP)

RTCC MFD > UTI > REF > VES: LM, Aquarius > OPT: P30 (Heads Down)

Uplink MCC-7 REFSMMAT (desired)

RTCC MFD > UPL > DIS > 23 > CUR 2

Entry REFSMMAT (desired)

First, calculate Earth Crescent REFSMMAT and uplink to CSM (actual)

Calculate Entry REFSMMAT

RTCC MFD > UTI > REF > VES: CSM, Odyssey > OPT: Lunar Entry

Uplink Entry REFSMMAT (desired)

RTCC MFD > UPL > DIS > 12 > CUR 2

Entry PAD (final)

First, calculate Entry REFSMMAT (desired)

RTCC MFD > PAD > ENP > OPT: Lunar Entry PAD > CLC (Verify P 0.05G $\approx$ 152)

